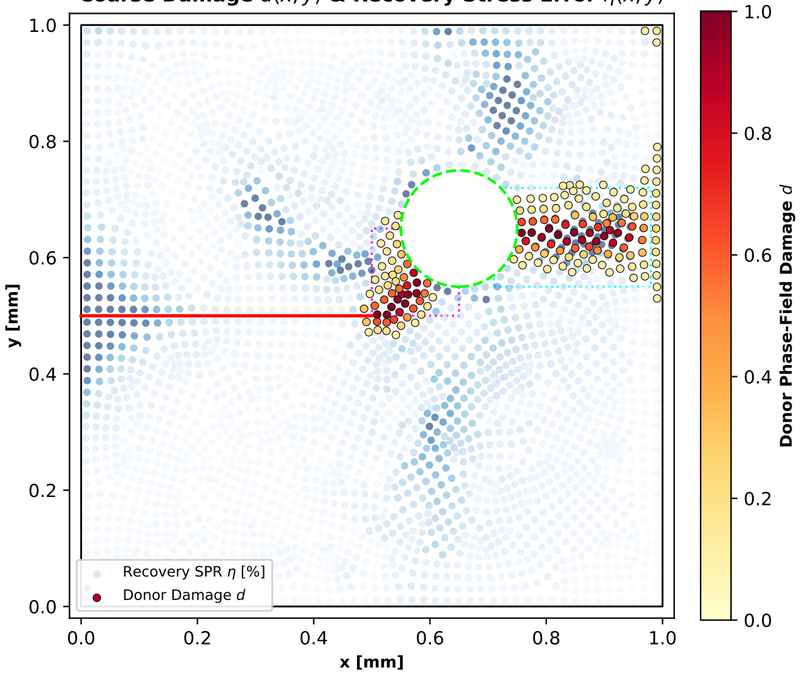
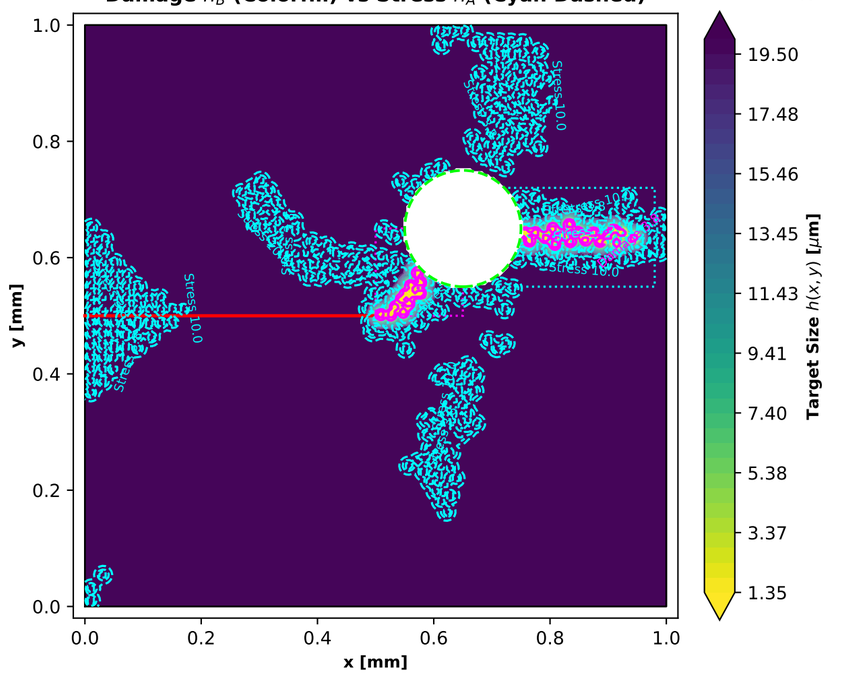
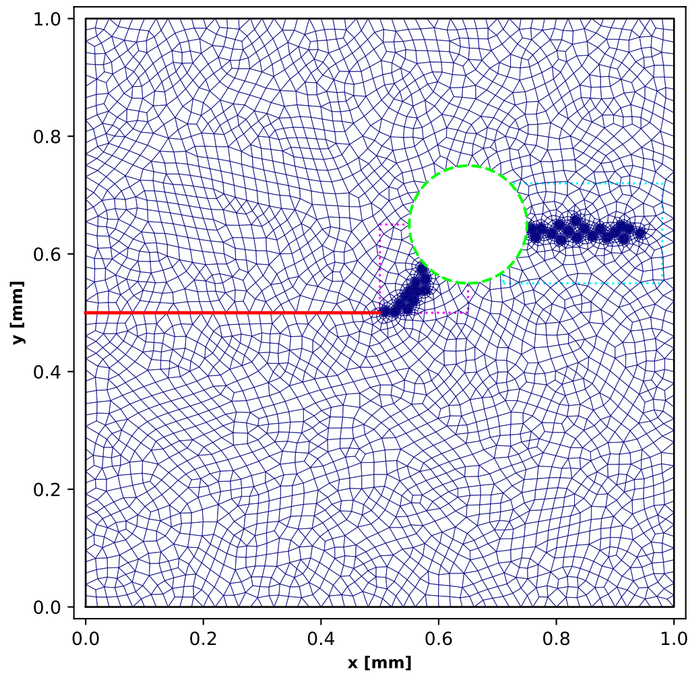
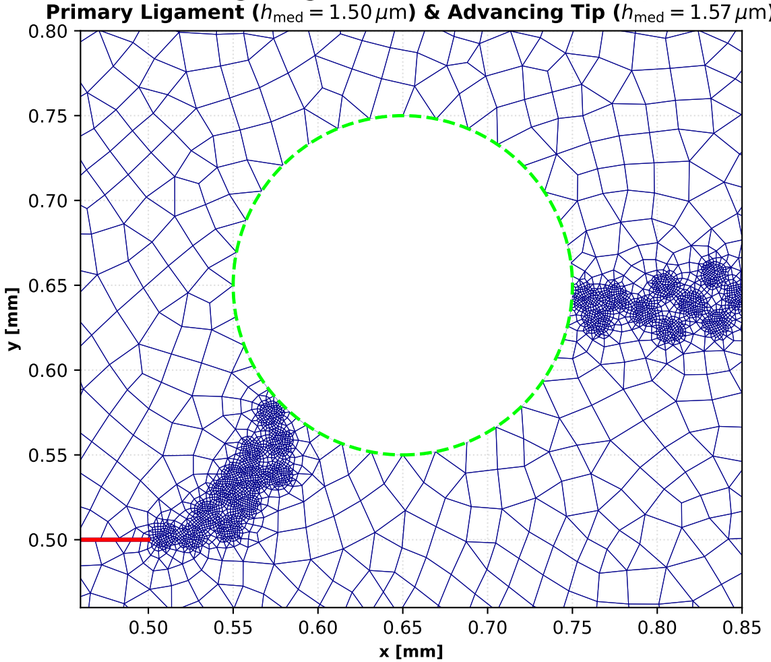
TASK F1668: CONTROLLED THREE-WAY ADAPTIVE REMESHING STUDY

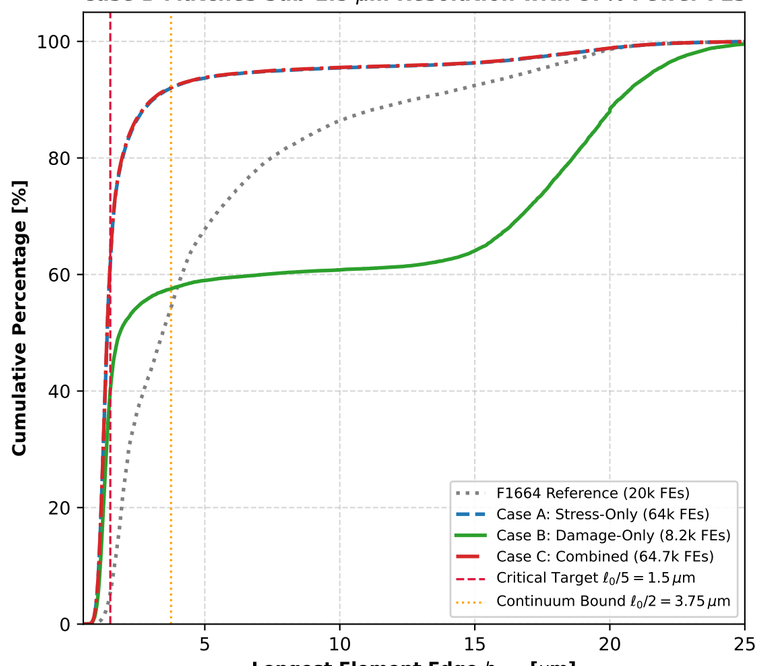
Mode-I Plate with Off-Axis Hole | Authentic Coarse Donor & Recovery Fields | Identical Geometry & Mesher











(f) Master Three-Way Scientific Evaluation Matrix
Case B Wins Efficiency-Resolution Trade-Off (8.2k vs 64.7k Elements)

Evaluation Metric	F1664 Ref	Case A (Stress	Case B (Damag	Case C (Hybrid
Total Elements	20,213	64,032	**8,237**	64,746
Pure Quads %	97.18%	78.03%	**77.67%**	78.09%
Non-Pos Jacobians	0	0	**0**	0
Distorted Elements	0	3	**0**	2
Global Sub-1.5 um	273 (1.4%)	27,554 (43.0%)	**2,299 (27.9%)**	28,097 (43.4%)
Ligament Median h	4.684 um	1.490 um	**1.501 um**	1.491 um
Ligament <=1.5 um	0 (0.0%)	3,289 (51.6%)	**904 (49.8%)**	3,270 (51.4%)
Secondary Med h	2.091 um	1.532 um	**1.570 um**	1.516 um
Secondary <=1.5 um	271 (3.7%)	5,095 (45.1%)	**1,347 (41.9%)**	5,662 (47.6%)
Far-Field Median h	5.530 um	1.570 um	**20.609 um**	1.573 um
Far-Field <=1.5 um	2 (0.0%)	19,170 (41.3%)	**48 (1.5%)**	19,165 (41.2%)
Far-Field Bloat	Low	Severe Bloat	**ELIMINATED**	Severe Bloat
Production FEs	60,639	192,096	**24,711 (Lean)**	194,238